Xuefeng Hu

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Member of Technical Staff | World Model https://xuefenghu.me

Research Focus

I develop world models and multimodal large language models. My recent work focuses on generalizable memory mechanisms for world models. During my Ph.D., I worked on domain adaptation for large-scale vision–language models, proposing methods for semi-supervised learning, source-free adaptation, and test-time adaptation in low-resource and distribution-shifted settings.

Education

08/2018 – 12/2023 **University of Southern California**
Los Angeles, CA Ph.D. in Computer Science
Dissertation: Adapt Pre-trained Representation Towards Downstream Tasks
Committee: **Ram Nevatia**, Aram Galstyan and Keith Jenkins

09/2017 – 08/2018 **Columbia University**
New York, NY M.S. in Computer Science
Advisor: Shih-Fu Chang

09/2015 – 05/2017 **University of Michigan, Ann Arbor**
Ann Arbor, MI B.S.E. in Computer Science
Advisor: Jia Deng

09/2013 – 08/2017 **Shanghai Jiao Tong University**
Shanghai, China B.S.E. in Electrical and Computer Engineering

Professional Experience

07/2026 – Current
Paris, France
Member of Technical Staff

Advanced Machine Intelligence

- Research memory mechanisms and mental-model representations for world models.
- Contribute to core training infrastructure, including linear-attention kernels, for efficient large-scale training.
- Contribute to data curation and evaluation methodology for world model research.

02/2024 – 07/2026
San Jose, CA
Staff Research Scientist

ByteDance / TikTok Inc.

- Conducted fundamental research on multimodal large language models (vision-language understanding, generation, and reasoning), publishing openly at peer-reviewed venues including NeurIPS, EMNLP, and CVPR.
- Publicly released open-source datasets and research models to the research community, including InfiMM-WebMath-40B (16K+ Hugging Face downloads, featured in HF Daily Papers) and DiCo (NeurIPS 2025 Spotlight).
- Co-authored academic papers on unified vision-language modeling, diffusion architectures, and multimodal reasoning, several accepted at top-tier international AI conferences.
- Mentored research interns on collaborative, publication-oriented projects.

05/2022 – 08/2022
Seattle, WA
Research Intern

Amazon Lab126

- Developed **ReCLIP**, a robust source-free unsupervised adaptation algorithm that improves CLIP's average zero-shot accuracy by **5.11%** across **22 classification benchmarks** without additional supervision.
- Presented the work as an oral at **WACV 2024**, where it was selected as a **Best Paper Award Finalist**.

05/2021 – 08/2021
Boston, MA
Research Intern

Meta – AI Applied Research

- Developed a novel **Test-Time Adaptation** method enabling pre-trained models to adapt to shifted test distributions in a fully online setting.
- Achieved **significant performance gains** over state-of-the-art (SOTA) methods across multiple challenging real-world settings.

05/2020 – 12/2020
Boston, MA
Research Intern

Facebook – AI Applied Research

- Proposed a **Part-level SimCLR** framework for weakly-supervised fine-grained anomaly detection, introducing novel contrastive learning techniques.
- Achieved **state-of-the-art (SOTA)** performance on multiple anomaly detection benchmarks.

Selected Publications

Ai, Y., Han, J., Zhuang, S., Mao, W., **Hu, X.**, Yang, Z., Yang, Z., Huang, H., Yue, X., & Chen, H. (2026). BitDance: Scaling Autoregressive Generative Models with Binary Tokens. Advances in Neural Information Processing Systems. **NeurIPS 2026**.

Chen, L., Wei, M., **Hu, X.**, & Liu, W. (2026). Intervention-Certified Amplification: A Validity Gate for σ-Weighted Post-Training. In Findings of the Association for Computational Linguistics. **EMNLP 2026 Findings**.

Wang, J.*, Wang, X.*, Pan, J.*, **Hu, X.***, Li, F., Sun, J., Deng, C., Chen, Z., Chen, Y., Tian, K., Gwilliam, M., Chen, H., Guan, D., Xu, K., Huang, W., Wu, Z., Fan, H., Jiang, Y.-G., & Yang, Z. (2026). ARM: An AutoRegressive Large Multimodal Model with Unified Discrete Representations. **Technical Report**, arXiv:2606.11188. * Co-First Author.

Gwilliam, M., Wang, X., **Hu, X.**, & Yang, Z. (2026). Implicit Neural Representation Facilitates Unified Universal Vision Encoding. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. **CVPR 2026**.

Ai, Y., Fan, Q., **Hu, X.**, Yang, Z., He, R., & Huang, H. (2025). DiCo: Revitalizing ConvNets for Scalable and Efficient Diffusion Modeling. Advances in Neural Information Processing Systems. **NeurIPS 2025 Spotlight**.

Han, X.*, Jian, Y.*, **Hu, X.***, Liu, H.*, Wang, Y.*, Fan, Q., Ai, Y., Huang, H., He, R., Yang, Z., & You, Q. (2025). InfiMM-WebMath-40B: Advancing Multimodal Pre-Training for Enhanced Mathematical Reasoning. In Findings of the Association for Computational Linguistics. **EMNLP 2025 Findings**. * Co-First Author.

Zheng, Z., Wei, J., **Hu, X.**, Zhu, H., & Nevatia, R. (2024). Large Language Models are Good Prompt Learners for Low-Shot Image Classification. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. **CVPR 2024**.

Hu, X., Zhang, K., Xia, L., Chen, A., Luo, J., Wang, K., Qiao, N., Zeng, X., Sun, Y., Sun, M., Kuo, C., & Nevatia, R. (2024). ReCLIP: Refine Contrastive Language Image Pre-Training with Source Free Domain Adaptation. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. **WACV 2024 Oral & Best Paper Award Finalist**.

Hu, Z., Yang, Z., **Hu, X.**, & Nevatia, R. (2021). SimPLE: Similar Pseudo Label Exploitation for Semi-Supervised Classification. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. **CVPR 2021**.

Hu, X., Zhang, Z., Jiang, Z., Chaudhuri, S., Yang, Z., & Nevatia, R. (2020). SPAN: Spatial Pyramid Attention Network for Image Manipulation Localization. In Proceedings of the European Conference on Computer Vision. **ECCV 2020**.

See my Google Scholar profile for the full publication list.

Skills

Programming
C/C++, Python, Java, SQL, Torch, JavaScript, R, Verilog

Frameworks
PyTorch, TensorFlow, PySpark

Others
MATLAB, Git, LaTeX, Lean

Awards

01/2024 **Best Paper Award Finalist**
IEEE/CVF Winter Conference on Applications of Computer Vision, 2024

03/2022 **"30 Under 30" Honoree, Class of 2022**
All America Chinese Youth Federation (AACYF)

03/2017 **James B. Angell Scholar**
University of Michigan

12/2016,04/2016, 12/2015 **Dean's List**
University of Michigan

04/2016,12/2015 **University Honors**
University of Michigan

08/2015 **Finalist (National Top 8 teams)**
TIC100 Smart City & IoT Business Challenge

08/2014 **University Merit Student (Top 3%)**
Shanghai Jiao Tong University

04/2014 **Outstanding Undergraduate Student Scholarship (Top 1%)**
Shanghai Jiao Tong University

12/2013 **Gold Medal**
University of Michigan-Shanghai Jiao Tong University Joint Institute 2013 Winter Design Expo

11/2013 **Silver Medal**
University of Michigan-Shanghai Jiao Tong University Joint Institute 2013 Robot Racing Competition

10/2013 **Outstanding Freshman Scholarship (Top 1%)**
Shanghai Jiao Tong University

11/2012 **National First Prize**
China High School Mathematics Competition, 2012